

VASUNDHHARA SINGH KATOCH

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EDUCATION

Bachelor's in Engineering, Computer Science

SSN College of Engineering, Chennai

Aug 2019 - May 2023

8.37 cgpa

WORK EXPERIENCE

AI Innovation Specialist- Product Management, IgniteTech

June 2025 - Present

- Contributed to **post-acquisition strategy and roadmap** for **modernizing a multi-tenant enterprise engagement platform** serving 250+ customers across 27 social channels, and processing 100M+ conversations annually delivering the **MVP in 3 months**.
- Applied deep technical fluency as an engineer to **co-own architecture decisions, API decisions, technical documentation**, build ready **PRD creation, and cross-functional execution** across Engineering, Design, and QA.
- Reduced app login time from 43s to 3s and a 7x smaller main bundle (5.2 MB → 727 KB), reducing customer deployment time by 75%, infrastructure costs by 80%, message ingestion latency from minutes to 195ms P50 (validated at 20K msgs/hr, zero errors), and ingestion engine deploy time 30x (30 min → <60s rolling updates).
- Designed scalable microservices, APIs, and event-driven architectures using Kafka, MySQL, MongoDB, Elasticsearch, and Redis, supporting real-time processing at a scale of 150M+ events per day.
- Led the AI product line**- 10+ shipped capabilities across 6 model providers with per-tenant controls, feature-level rollout, and token-cost governance, converting agent engagement and publishing into AI-assisted workflows, **cutting agent response drafting from 3-4 minutes to ~1 minute (4x)**.
- Drove **enterprise readiness- multi-tenancy, SSO/OAuth, RBAC, tenant isolation, audit trails, and data residency** required by enterprise buyers.
- Instrumented **product metrics (adoption, AI usage, cost-per-interaction)** via **Grafana and Langfuse**, reviewing them weekly to guide roadmap decisions.
- Delivered pilot launches across teams- driving customer engagement and product transformation, and ran post pilot retros to feed the next cycle, with emphasis on rapidly validating, rebuilding, and launching, in a phased migration from legacy systems.

Product Engineer | AI and Backend, Hushh AI

Jan 2024 - May 2025

- Product Management & Strategy**
 - Led end-to-end PM for four products (**B2B SaaS & B2C**), driving **market research, customer interviews, product requirements, user journeys** and **GTM execution**.
 - Conducted 50+ customer interviews to refine **Product Market Fit (PMF)**.
 - Reduced Customer Acquisition Cost (CAC) by 60%** by restructuring the Go-to-Market (GTM) strategy and streamlining acquisition channels.
 - Led product development across **iOS, Android, and Web**, ensuring seamless cross-platform user experiences.
 - Configured event tracking for **Customer Lifecycle Management (CLM)** using **Clevertap**, optimizing engagement and retention strategies.
 - Implemented product analytics systems using **Google Analytics**, enabling **data-driven insights** for **user behavior tracking and funnel optimization**.
- Partnerships & Stakeholder Engagement**
 - Secured **design partnerships** with 5+ leading brands.
 - Delivered **technical demos** showcasing product capabilities to stakeholders and partners.
- Market Representation**
 - Represented the company at various international events, including **Abu Dhabi Finance Week 2024**.
- Cross-Functional Leadership**
 - Coordinated with **project management team** to establish **Agile** processes from scratch, and **led a 12-member engineering team** reducing release cycles from 6 → 3 weeks, cutting missed deadlines by 40% and improving sprint velocity by 30%, tracked using **tailored KPI metrics**.
 - Bridged engineering, product, project and leadership teams and conducted quarterly team reviews.
- Technical Execution**
 - Oversaw **technical documentation/API guides** and collaborated on product architecture.
 - Worked on **technical architectures** for multiple products, collaborating directly with founders on product strategy.

AI and backend engineer, Hushh AI

Sep 2023 - Jan 2024

- Lead research and development of a **Personal Knowledge Management (PKM)** system, integrating structured and unstructured data to create an AI-driven personalised chatbot. Managing the **full development lifecycle**, including system architecture, testing, and deployment. Explored multiple approaches to enhance data retrieval and personalization:
 - LLM-generated GraphQL queries for extracting relevant user data. (Tech stack: **GraphQL, LangChain, LLaMA models, Hugging Face, Ollama, FastAPI**)
 - LightRAG-based retrieval system. (Tech stack: **FastAPI, LightRAG implementation based on research paper**)
 - GraphRAG-based retrieval system**. (Tech stack used: **FastAPI, Neo4j (A Graph DB)**)
 - Spearheading deployment strategy, considering **Vertex AI, Docker, self-hosted GPUs, hugging face, GCP**, for **efficient costing and model performance**.
- Developed and implemented an end-to-end **AI hiring assistant** using **OpenAI, Langchain, Hugging Face libraries, Pandas, PDF Parsing techniques, RAG** enhancing the recruitment process with automated candidate screening.
- Designed comprehensive database schemas and built the architecture to enable seamless data flow across multiple products.
- Developed robust Developer APIs using **FastAPI**, with a secure onboarding. Also leveraged **FastAPI realtime processes, webhooks and notifications** using **Firebase Cloud Messaging**.

SKILLS

Technical Product Management: System Architecture & Design, API Contract Definition, Data Modeling, Multi-Tenant System Design, Platform Migration Strategy, AI Platform Architecture, Real-Time Systems, Asynchronous Data Pipelines

Project Management: Jira, Confluence, Github, Agile, Scrum, Kanban, Roadmap Planning, Sprint Planning & Execution

Analytics & Data: Google Analytics, Grafana, Langfuse, SQL

Design & Research: Figma, User Research, User Journey Design, Prototyping,

AI tools: Cursor, Claude, Claude Code, Gemini, Perplexity, CodeRabbit

Languages: Python, C++, JavaScript, SQL

Databases: PostgreSQL (Supabase), Firebase, MySQL, MongoDB, DynamoDB, Elasticsearch, Redis, Neo4j, Vector Databases

Areas of Interest: Generative AI, RAG, NLP, Graph Databases, API Development